

Cole R. Pearson

Biomedical engineering graduate with laboratory experience, management experience, strong teamwork skills, problem-solving abilities, and well-developed communication.

PROFESSIONAL EXPERIENCE

STABILITY BIOLOGICS – Quality Control Supervisor

July 2024 – April 2026

- Performed installation, operational, and performance qualification (IQ/OQ/PQ), completed equipment calibrations, and produced new maintenance protocols and validation reports for laboratory equipment
- Wrote non-conformance reports (NCRs), led root cause analysis (RCA), and executed the resulting corrective and preventive action (CAPA)
- Applied design control processes and drove continuous improvement for human biological material (tissue allografts) manufacturing in a sterile environment
- Directed quality control and manufacturing operations at a certified tissue bank, ensuring all activities stayed fully compliant with 21 CFR 1271, 21 CFR 820, AATB Standards, and ISO 13485
- Implemented Six Sigma lean principles in an FDA-regulated manufacturing setting and upheld current good manufacturing practices (cGMP)
- Managed and organized batch records and laboratory information, and consolidated datasets to monitor key trends and strengthen root cause analysis
- Conducted laboratory procedures and processed bone and birth tissues using aseptic techniques
- Performed routine in-process and stability testing on donor tissues and finished products
- Planned and executed environmental monitoring activities, including plating and particle counts
- Built and deployed automated product-labeling workflows using BarTender
- Trained new and current personnel to maintain ongoing competency and up-to-date knowledge of standard operating protocols (SOPs)
- Managed and led a team of up to 20 quality control and manufacturing personnel, achieving 50% productivity gains while upholding quality standards

TEXAS A&M UNIVERSITY – Data Entry Specialist

September 2022 – September 2023

- Collaborated with a team to produce high-quality datasets used to draw data-driven conclusions
- Coordinated with a team of engineers to define project scope and establish data constraints
- Converted qualitative observations into objective, quantitative data

TEXAS A&M UNIVERSITY – Biomedical Research Assistant

February 2023 – May 2023

- Conducted nuclear magnetic resonance (NMR) and ultraviolet/visible (UV/Vis) spectroscopy to characterize materials
- Performed Alamar Blue and Hematoxylin and Eosin assays to support histological studies
- Prepared reagents for polymer synthesis and cultured mammalian cells (specifically cardiomyocytes) using aseptic techniques
- Gained working familiarity with LabVIEW and laboratory automation
- Applied statistical methods to experimental results and reported findings to the principal investigator
- Conducted chemistry research to establish a synthesis protocol for a reversible crosslinking polymer gel structure, then synthesized 3D electro-spun polymer scaffolds using reversible crosslinking chemistry

TEXAS A&M UNIVERSITY – Mathematics Teaching Fellow

September 2019 – April 2020

- Taught engineering students' problem-solving methods relevant to engineering calculus

UNIVERSITY OF TEXAS AT SAN ANTONIO – Biomedical Research Assistant

June 2019 – August 2019

- Wrote a project abstract on electro-spun polymer scaffolds for tracheal wounds
- Conducted polymer absorption, degradation, adhesion, and mechanical testing
- Performed scanning electron microscopy and mammalian cell culturing (specifically T lymphocytes) using aseptic technique and pipetting skills
- Used statistical methods to analyze experimental results and summarized them in the project abstract
- Presented research findings at the 2019 Biomedical Engineering Society Conference

ACCENTURE FEDERAL – Salesforce Development Intern

February 2018 - May 2018

- Developed and tested interactive web applications with a team of qualified developers
- Worked with databases and web programming in SQL Server and Java
- Created web applications using HTML, CSS, and JavaScript

EDUCATION

TEXAS A&M UNIVERSITY

BS Biomedical Engineering, 2023

- Hands on laboratory educational experience with high performance liquid chromatography (HPLC), enzyme-linked immunosorbent assay (ELISA), quantitative reverse transcription polymerase chain reaction (qRT-PCR), western blot, titration, gel electrophoresis, and computational genomics
- Basic programming skills for data analysis using Python and R for statistics

TAMU Capstone Design Innovation Award

Computerized System or Software Design

- Created an initial prototype for a novel computerized ureteroscopy device
- Interviewed industry experts and conducted market research to establish user needs and project scope for a 510k pathway medical device
- Coded prototype software in C++ and soldered micro-electromechanical components
- Fabricated prototype hardware components using fused deposition 3D printing
- Applied design iteration methods such as design failure modes and effects analysis (DFMEA) and established a design process control map to achieve verification and validation
- Authored a complete design history file (DHF) compliant with FDA requirements

Relevant Coursework:

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| • Physiology for Bioengineers | • Differential Equations |
| • Drug Delivery | • Biosolid Mechanics |
| • Biosystems Mass and Energy Transfer | • Biomolecular Engineering |
| • Statistics for Bioengineers | • Molecular and Cellular Biomechanics |
| • Biomedical Nanotechnology | • Computational Genomics |
| • Medical Imaging | • Medical Device Design |

BUSINESS CAREERS HIGH SCHOOL

Information Technology Track, 2018

- Certified in Microsoft Word, PowerPoint, and Excel
- Certified in Adobe Dreamweaver
- Achieved Internet Core Competency Certification (IC₃)

Programming Languages:

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| • C++ | • Python | • HTML |
| • LabVIEW | • R for Statistics | • CSS |
| • SQL | • PowerShell | • JavaScript |
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